

Convex Optimization

{Raj: Appendix B of "ML Foundations"}

Def A set $X \subseteq \mathbb{R}^n$ is convex if, for all $x, y \in X$,
$$\alpha x + (1-\alpha)y \in X \quad \forall \alpha \in [0,1].$$

In other words, a set is convex if the line segment joining any two points in the set lies within the set.

Eg: Let A be an $m \times n$ matrix, b an $m \times 1$ vector.

$\{x \in \mathbb{R}^n : Ax + b \leq 0\}$ is convex
pointwise

Note: The set is the intersection of half-spaces.
Also called a polyhedral set.

Operations that preserve Convexity: Lemma: Say C_1 & C_2 are convex sets. Then

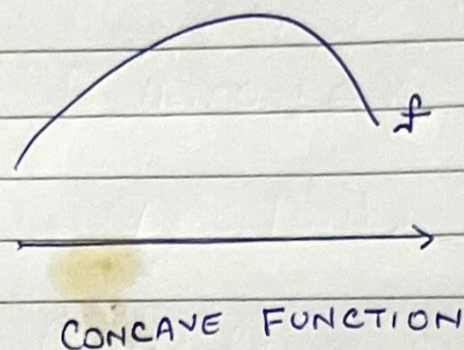
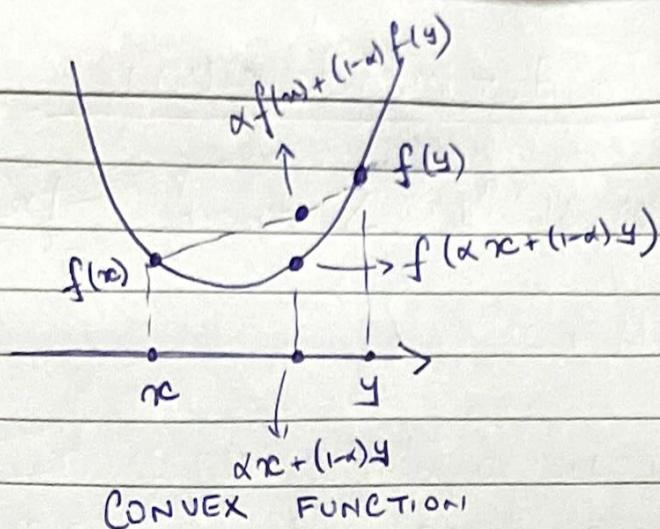
- $C_1 \cap C_2$ is convex
- $C_1 \times C_2$ is convex
- $C_1 + C_2 := \{x_1 + x_2 : x_1 \in C_1, x_2 \in C_2\}$ is convex

↳ Exercise

Def Say X is convex. A function $f: X \rightarrow \mathbb{R}$ is convex if, for all $x, y \in X$, $\alpha \in [0,1]$,

$$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y). \quad - (1)$$

↳ Graph of function lies below any chord joining 2 points on the graph.



- Def: • f is strictly convex if ① is strict whenever $x \neq y, \alpha \in (0,1)$.
- f is concave $\Leftrightarrow -f$ is convex.
 - f is strictly concave $\Leftrightarrow -f$ is strictly convex.

Theorem: Say $f: X \rightarrow \mathbb{R}$ is differentiable & X is convex.
Then f is convex iff

$$f(y) \geq \underbrace{f(x) + (y-x)^T \nabla f(x)}_{\text{tangent hyperplane at } x} \quad \forall x, y \in X.$$

Moral: A convex function is bounded from below by its tangent hyperplanes.

Theorem Say $f: X \rightarrow \mathbb{R}$ is twice differentiable, and X is convex. Then f is convex (resp. strictly convex) iff $\nabla^2 f$ is positive semidefinite (resp. positive definite) $\forall x$.

Eg For a positive semidefinite matrix Σ , the function $x^T \Sigma x$ is convex over $\mathbb{R}^n$.

Eg Any norm ^{$\|\cdot\|$} over a convex set X is convex, since by triangle inequality,

$$\|\alpha x + (1-\alpha)y\| \leq \|\alpha x\| + \|(1-\alpha)y\| = \alpha\|x\| + (1-\alpha)\|y\|.$$

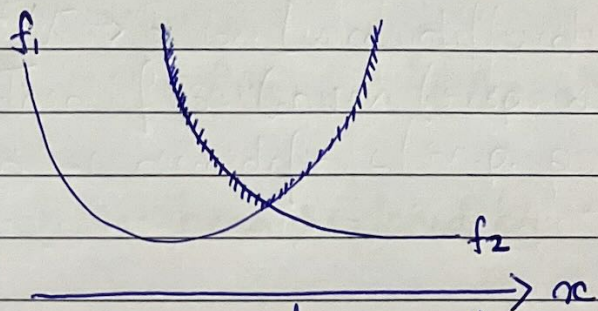
Theorem (Jensen's inequality) Let X be a random variable taking values in a convex set X . Then for a convex function f over X ,

$$f(E[X]) \leq E[f(X)].$$

Lemma Let $\{f_i\}_{i \in I}$ be a family of convex functions defined on a convex set X . Then the pointwise supremum, given by

$$f(x) = \sup_{i \in I} f_i(x)$$

is convex.



Moral: Pointwise supremum of convex functions is convex.

Now, suppose that X is convex; f is convex over X ,
 $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$

We consider the following convex optimization with linear constraints:

$$\begin{aligned} \min_{x \in X} \quad & \underbrace{f(x)}_{\text{objective}} \quad (P) \\ \text{subject to:} \quad & Ax \leq b \quad \text{constraints} \end{aligned}$$

- This is called the primal problem (P)
- Linear constraint can be replaced by convex constraints $g_i(x) \leq 0$, $1 \leq i \leq m$, where $g_i(\cdot)$ are convex over X .
 $\hookrightarrow$ see handout
- Effectively (P) minimizes f over $\underbrace{X \cap \{x: Ax \leq b\}}_{\text{Convex set}}$
- Let p^* denote the optimal value of (P).

the Lagrangian

Now, define $L: X \times \mathbb{R}_+^m \rightarrow \mathbb{R}$ as follows:

$$\begin{aligned} L(x, \alpha) &= f(x) + \sum_{i=1}^m \alpha_i (a_i x - b_i) \\ &= f(x) + \alpha^T (Ax - b) \end{aligned} \quad \left\{ A = \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix} \right.$$

Note: Constraints have been replaced by "violation penalties"

Def: The (Lagrange) dual function $F: \mathbb{R}_+^m \rightarrow \mathbb{R}$ is defined as:

$$F(\alpha) = \inf_{x \in X} L(x, \alpha)$$

Note: • F is concave { pointwise inf. of linear functions

• $F(\alpha) \leq p^* \left\{ \begin{array}{l} \text{For feasible } x, \\ f(x) + \sum \alpha_i (a_i x - b_i) \leq f(x) \\ \Rightarrow \text{Inf of LHS over } X \leq \text{Inf of RHS} \\ \text{over feasible points} \end{array} \right.$

The Dual optimization associated with (P) is:

$$\max_{\alpha \in \mathbb{R}_+^m} F(\alpha) \quad (D)$$

• Let d^* denote optimal value of (D). We know $d^* \leq p^*$

so long as feasible set is non-empty)
• Turns out (~~so long as set of feasible~~ that $d^* = p^*$ (strong duality; no duality gap)

Note: When dealing with convex constraints, we need a bit more for strong duality; see Slater's conditions.

Theorem x^* solves (P) if and only if $\exists \alpha^* \geq 0$ s.t.

KKT
Conditions

$$\nabla_x L(x^*, \alpha^*) = \nabla_x f(x^*) + \sum \alpha_i^* a_i^T = 0$$

$$\nabla_{\alpha} L(x^*, \alpha^*) = Ax^* - b \leq 0$$

$$\alpha^{*T} (Ax^* - b) = 0$$

In this case, α^* solves (D) & (x^*, α^*) is a saddle point of L .

Moral: • Under mild conditions, solving (P) is equivalent to solving (D). Often, one may be more efficient/insightful to work with.

- Solving either is equivalent to finding (x^*, α^*) satisfying KKT conditions.